



TECHNICAL CASE STUDY

Cross-Border Payment Platform Integration

Position: Solutions Architect

DOKU — PT Nusa Satu Inti Artha (Indonesia)

senangPay — Simplepay Gateway Sdn Bhd (Malaysia)

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- You will not share this document with other candidates, current or former DOKU or senangPay employees, or any third party.
- You will not include confidential information belonging to your current or former employers in your submission. If your best example is confidential, describe the shape of the problem, not the data.
- You may retain a copy of your own submission for your personal records.

1 How to Read This Document

We are not looking for a correct answer. The scenario below deliberately contains requirements that contradict one another and information that is missing. That is part of the problem.

What we assess is whether you:

- identify what does not add up, and say so;
- decide not to build certain things, and explain why;
- distinguish an architecture decision from a decision that belongs to legal, treasury, or a regulator;
- commit to a defensible position on incomplete information.

Use of AI tools is permitted. We use them too. We ask only that you disclose, in three lines at the end of your submission, which parts you produced with AI assistance and how you verified them. Your submission is discussed with you in a live session, so a document you do not understand will become apparent quickly. Honest disclosure carries no penalty.

2 Corporate Context

DOKU Group operates two payment businesses under two separate legal entities, in two jurisdictions, under two regulators.

	DOKU Indonesia	senangPay Malaysia
Legal entity	PT Nusa Satu Inti Artha	Simplepay Gateway Sdn Bhd
Regulator	Bank Indonesia — PJP Category 1 Licence	Bank Negara Malaysia — registered Merchant Acquiring Services provider
Scheme status	Five BI licences: payment gateway, disbursement/remittance, e-money, e-wallet, QRIS	Registered with Mastercard International as Payment Facilitator (APAC)
Merchants	150,000+, enterprise and SME	15,000+, predominantly SME
Currency	IDR	MYR
Operating timezone	WIB (UTC+7)	MYT (UTC+8)
Certification	PCI DSS Level 1 — must be maintained	

senangPay was acquired in 2022. Three years later the two platforms remain entirely separate: separate merchant identifiers, separate dashboards, separate ledgers, separate settlement cycles, separate support queues. Nothing is integrated. You have just been appointed Solutions Architect for the group programme that is expected to change this.

3 The Systems as They Stand Today

All figures below are fictional and simplified for assessment purposes.

3.1 DOKU Indonesia — “PG-Core”

Application	Java 11 / Spring Boot monolith, nine years old
Database	MySQL 8, one primary and two read replicas, single region
Persistence	Hibernate, used throughout — including on the authorisation path
Messaging	Kafka, adopted by one of three squads; the rest use synchronous HTTP and cron
Infrastructure	On-premise data centre Jakarta, disaster recovery site Surabaya. RPO 15 minutes, RTO 4 hours
Integrations	23 acquirers and banks, each with a different message format and cut-off time
Engineering	14 engineers across 3 squads, 2 SRE

Average throughput	350 TPS
Daily peak	1,200 TPS
Harbolnas (12.12) peak	6,000 TPS sustained for ~90 minutes; burst 9,500 TPS
Authorisation p99, end to end	1,850 ms
Largest acquirer p99 (not under our control)	450 ms; timeout 5,000 ms; 0.4% of requests time out
Merchant callbacks	0.3% are lost or delivered twice
Stuck / pending transactions	0.08% of volume — approximately 4,800 items per day, resolved manually
Daily reconciliation completes	D+1 at 11:00 WIB

3.2 senangPay Malaysia — “SP-Gateway”

Application	PHP 8 / Laravel, built 2015, incrementally extended
Database	PostgreSQL 14, single primary with streaming replica
Infrastructure	Public cloud. Newer workloads in the Malaysia region; several legacy workloads still run from the Singapore region
Payment methods	Cards (via Payment Facilitator arrangement), FPX internet banking, major Malaysian e-wallets
FPX characteristics	Scheme-defined cut-off and batch settlement. No instant reversal; refunds are a separate credit transfer
Engineering	10 engineers, 1 SRE (total company headcount 45)
Merchant onboarding	Largely manual; KYC performed in-country

Average throughput	90 TPS
Peak	400 TPS
Authorisation p99	850 ms
Daily reconciliation completes	D+1 at 09:00 MYT

3.3 What connects them

Nothing. There is no shared identity, no shared ledger, no shared merchant record, and no common API. A merchant operating in both countries holds two accounts, integrates twice, and reconciles twice.

4 What the Stakeholders Are Asking For

The following requests were made to you in your first two weeks. They are reproduced as stated. Beneath each, we set out what your submission must address. You need not answer under these headings — ensure only that a reader can find each answer.

4.1 Group Chief Executive Officer

“One global API within nine months. An Indonesian merchant should be able to accept MYR, and a Malaysian merchant should be able to accept IDR. One onboarding, one dashboard, one integration. And obviously exactly-once processing — no double charges, ever.”

Your submission must address:

- Which parts of this request can be delivered, which cannot, and which are not architecture decisions at all. Where a request is impossible, explain why and state what you offer instead.
- How you guarantee that no money is created, lost, or duplicated when the network fails mid-transaction. Address idempotency, the ledger, and how a transaction moves between states.
- IDR and MYR do not share the same minor unit convention. At what moment is the foreign exchange rate captured — authorisation, capture, or settlement? If a payment is refunded three weeks later, at which rate? Who carries the difference?

4.2 Group Chief Financial Officer

“Put both entities' ledgers in a single database so consolidated reporting stops being a monthly fire drill. And net the balances between the two entities rather than physically moving cash back and forth every settlement cycle.”

Your submission must address:

- What must never cross the boundary between the two entities, and what may.
- How your architecture makes the impermissible case difficult to build by accident, rather than merely documented as forbidden.

4.3 Head of Data (Kuala Lumpur)

“Replicate the Indonesian transaction database into our analytics warehouse in KL, including the full card PAN. Our fraud team needs to trace patterns per card across both markets. It is just a read replica — should be a small piece of work.”

Your submission must address:

- How you respond to this request.
- If you decline, how the fraud team still gets what it needs.

4.4 Group Head of Operations

“Reconciliation must close before 06:00. But FPX has its own cut-off, the Jakarta team works WIB and the KL team works MYT, so nobody agrees on what ‘yesterday’ means. And we cannot keep clearing 4,800 manual items a day in Jakarta with six people.”

Your submission must address:

- Your definition of the business date, and why.
- How reconciliation closes before 06:00 given the FPX cut-off, and how the 4,800 daily manual items are reduced.
- What causes a transaction to become stuck in the first place.

4.5 VP Product

“Let’s just let our Indonesian merchants display a QRIS code that Malaysian tourists can scan with DuitNow, and vice versa. The regulators already announced cross-border QR, so this should mostly be a front-end change.”

Your submission must address:

- What you would need to establish before committing to cross-border QR acceptance.
- The first thing you would verify, and with whom.

4.6 Cutting across all five requests

Your submission must also address the following, which belong to no single stakeholder.

Failure modes.

Describe the behaviour of your system in each of the following. Name what degrades, what continues, and who is told.

- The largest Indonesian acquirer is unavailable for twenty minutes at the Harbolnas peak.
- The FPX settlement batch arrives three hours late.
- The MySQL primary fails while 4,000 transactions are in flight.

Scale and cost.

How does the platform absorb 6,000 TPS sustained within a +18% infrastructure budget? State what you would measure to prove your capacity assumptions — not merely which technology you would use.

Standards without coupling.

How is SNAP compliance satisfied without your internal domain model becoming locked to the shape of the regulator's API?

Organisation.

Fourteen engineers in Jakarta, ten in Kuala Lumpur, two timezones, two entities. How do the service boundaries you have drawn map onto the teams? What has to change in how they work?

Sequencing against a date you cannot move.

senangPay's Mastercard Payment Facilitator registration falls due for renewal in month seven, and material changes to merchant onboarding trigger a re-review. The Chief Executive's nine-month plan runs straight through it. How do you sequence the migration around this? If you choose to accept the risk instead of avoiding it, say so, state what you would do to contain it, and name who must approve that decision.

4.7 Where you have met this before

Payments architecture is not learned from documentation. Before you design, tell us what you already carry.

Your submission must address:

- Which of the requests above — 4.1 to 4.6 — resemble a problem you have actually faced in your career. Name the section number. If none do, say so plainly. "I have not faced this" is an acceptable answer and carries no penalty.
- For the closest one: what the system was, what your role in it was, and what the hardest part turned out to be. Note that the hardest part is rarely the part that looked hardest at the start.
- How it was resolved, who else had to agree, and what it cost — in time, in money, or in trust.
- One thing from that experience you would carry into this design, and one thing you would deliberately leave behind.

5 Constraints

Your design must remain compatible with the following. You are not expected to cite regulations. You are expected to know where a regulation changes the shape of an architecture.

5.1 Indonesia

- SNAP (Standar Nasional Open API Pembayaran) is mandatory for payment service providers. It standardises communication protocol, data format, authentication, authorisation, and encryption.
- QRIS scheme rules; five Bank Indonesia licences to preserve.
- Law No. 27 of 2022 on Personal Data Protection (UU PDP).
- Anti-money-laundering and counter-terrorism-financing obligations (APU-PPT); regulatory reporting to Bank Indonesia.
- Business Continuity Plan and Business Recovery Plan obligations.

5.2 Malaysia

- Bank Negara Malaysia registration as a Merchant Acquiring Services provider.
- Mastercard Payment Facilitator programme rules, including merchant onboarding and monitoring obligations.
- senangPay's Payment Facilitator registration falls due for renewal in month seven of the programme. Material changes to merchant onboarding trigger a re-review. The date is set by the scheme and cannot be moved.
- FPX scheme rules and settlement windows.
- Malaysian Personal Data Protection Act 2010, as amended.

5.3 Group

- PCI DSS Level 1 certification must be maintained without interruption.
- The two entities are legally distinct. Movements of funds and of personal data between them are regulated events, not internal transfers.
- Cross-border QR interoperability between the two countries operates through designated scheme arrangements. It is not something two payment service providers may wire together bilaterally.
- Additional infrastructure spend is capped at +18% year on year, group-wide. Headcount is frozen.
- The Chief Executive's nine-month expectation has been communicated to the board.

6 Your Deliverable

One document, in PDF or Markdown, containing the following.

#	Section
1	Assumptions and questions for DOKU. At least five that would change your design if answered differently.
2	Target architecture and a staged migration path. No big-bang rewrite of either platform.
3	One diagram. C4 level 2, component, or sequence. A legible photograph of a whiteboard is fine.
4	Non-goals. What you will not build or change in the first twelve months, and why.
5	One ADR for the decision most expensive to get wrong. Context / Decision / Alternatives rejected / Consequences, including the negative ones / How we would know it was wrong.
6	Note on AI assistance used.

6.1 Length

Six pages maximum, inclusive of the ADR, plus one diagram. Exceeding the limit reduces your score on Communication. Deciding what to cut is part of the work.

6.2 Time

This exercise is designed to take about three hours. Please return it within five working days of receipt.

We do not measure elapsed time, and we do not want you to spend a weekend on it. A submission that plainly consumed twelve hours is not a positive signal: the competency under test is reaching a defensible decision under constraint, not producing an exhaustive document. If you need longer for reasons of work, care, health, or timezone, tell your recruiter. It will not affect your assessment.

6.3 Technology

Any language, stack, or technology. There is no hidden preference. If you believe a component of either current system is the wrong choice for this problem, say so and demonstrate it.

A Appendix — Glossary

ADR	Architecture Decision Record. A short document capturing one decision, its alternatives, and its consequences.
APU-PPT	Indonesian anti-money-laundering and counter-terrorism-financing obligations.
BI	Bank Indonesia, the Indonesian central bank and payment system regulator.
BNM	Bank Negara Malaysia, the Malaysian central bank.
CDE	Cardholder Data Environment. The systems in scope for PCI DSS.
DuitNow	Malaysian national real-time payment and QR scheme, operated by PayNet.
FPX	Financial Process Exchange. Malaysian internet-banking payment scheme.
PAN	Primary Account Number. The long number printed on a payment card.
Payment Facilitator	A card-scheme role permitting an entity to onboard sub-merchants under its own acquiring relationship.
PCI DSS	Payment Card Industry Data Security Standard. Level 1 is the most stringent tier.
PJP	Penyedia Jasa Pembayaran — Payment Service Provider licensed by Bank Indonesia.
QRIS	Quick Response Code Indonesian Standard. The national QR payment standard.
SNAP	Standar Nasional Open API Pembayaran. Bank Indonesia's mandatory open API standard for payments.
TPS / p99	Transactions per second. The 99th-percentile latency: one request in a hundred is slower than this.
UU PDP	Law No. 27 of 2022 on Personal Data Protection (Indonesia).
WIB / MYT	Western Indonesian Time (UTC+7) / Malaysia Time (UTC+8).

B Appendix — Acknowledgement

Please complete, sign, and return this page together with your submission.

☐ I have read Section 0 of this document. I will not publish, forward, or otherwise share this case study or my submission with any third party.

☐ My submission contains no confidential information belonging to my current or former employers.

☐ I have disclosed any AI assistance used, as described in Section 1.

Full name:

Signature:

Date:

Questions about this case study should be directed to your recruiter, not to DOKU or senangPay engineering staff.